SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Dashboard Development Task

Course Code:	DSA0502	**Course Title:**	Query Processing for Data Science
CO Assessed:	CO5	**Assessment:**	Tool 3– Dashboard Development Task
Weightage:	25%	**Total Marks:**	
CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication			
Student Name:	Sania Herbert	**Reg. No.:**	192324062
Section / Batch:	2023/AI&DS	**Date of Submission:**	18.8.26
Faculty In-charge:	**Dr. B.Shyamala Gowri**	**Project Mode:**	Individual Submission

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: Sania

1. **Hospital Patient Analysis Dashboard**
   Build a dashboard using patient data containing age, blood pressure, cholesterol, BMI, treatment cost, and admission date. Present distributions, counts, and relationships between health-related variables using Matplotlib. Include time-based plots to analyze admissions and use lag/autocorrelation plots to identify recurring admission patterns.

Objective

The aim is to analyze hospital patient data using **Pandas and Matplotlib**. The dashboard will study patient demographics, health measurements, treatment costs, and admission patterns over time.

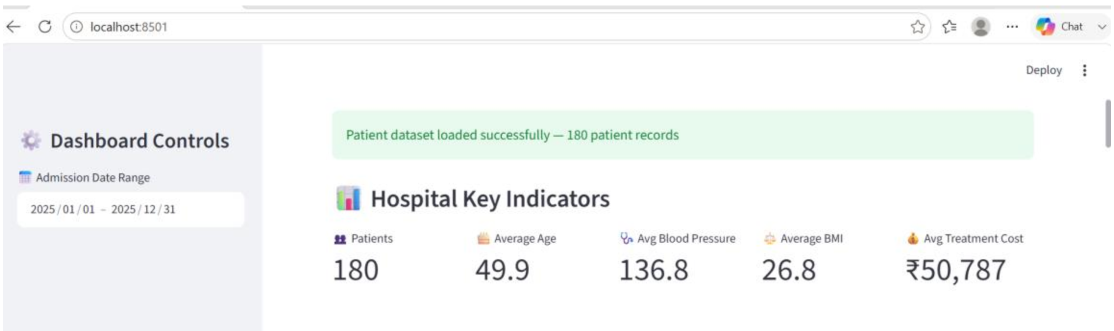
Dataset

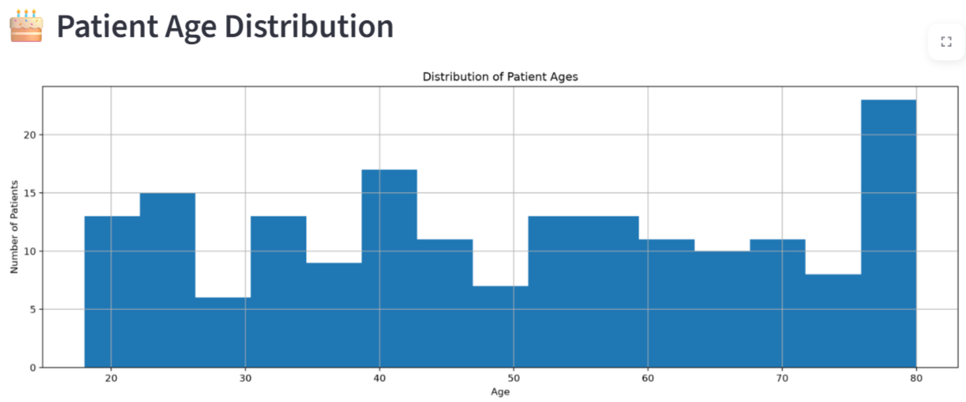
The dataset can contain:

Column	Description
Age	Patient's age
Blood_Pressure	Patient's blood pressure
Cholesterol	Cholesterol level
BMI	Body Mass Index
Treatment_Cost	Cost of treatment
Admission_Date	Date of hospital admission

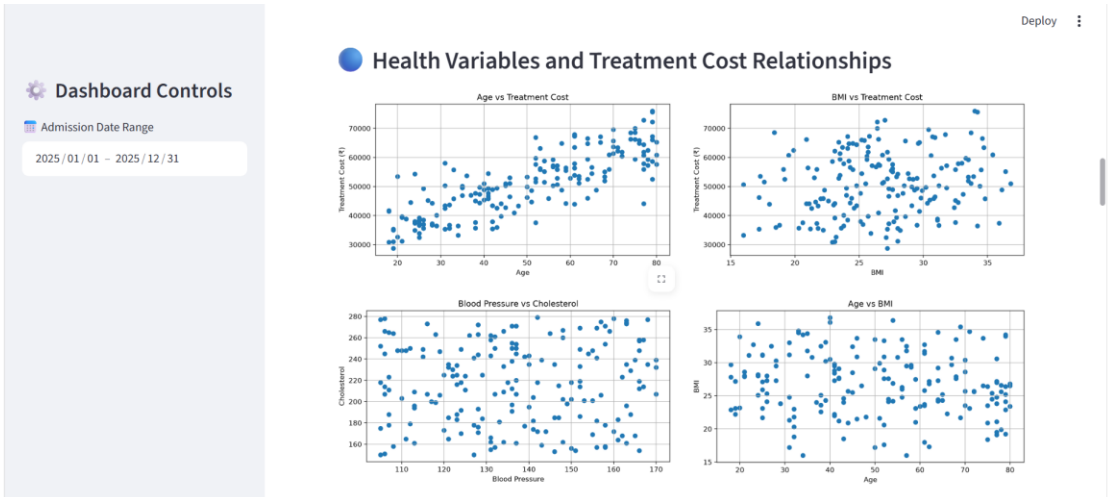
Age	Blood_Pressure	Cholesterol	BMI	Treatment_Cost	Admission_Date
25	118	180	22.5	15000	2025-01-05
42	135	220	27.8	32000	2025-01-08
56	145	245	31.2	48000	2025-01-15
35	125	195	24.6	21000	2025-02-02
68	155	260	33.5	65000	2025-02-10

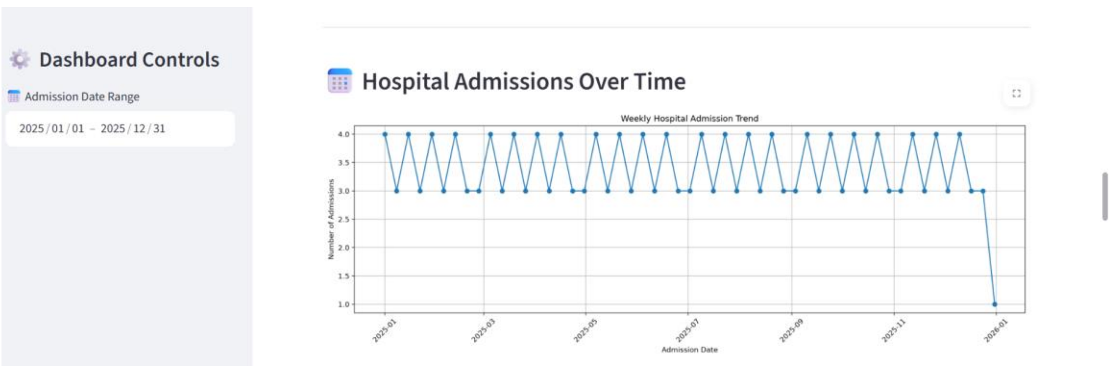

Dashboard Controls
Admission Date Range
2025 / 01 / 01 – 2025 / 12 / 31
Hospital Patient Analysis Dashboard
Patient Health, Cost and Admission Pattern Analysis
This interactive dashboard uses Pandas and Matplotlib to analyze patient demographics, health measurements, treatment costs and hospital admission patterns.
Techniques used:
Distributions
Count analysis
Scatter plots
Time-series analysis
Lag plot
Autocorrelation
Correlation analysis

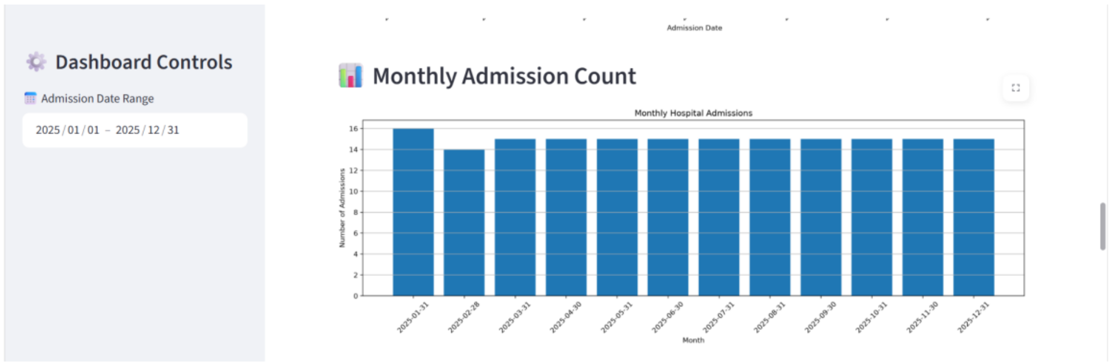

localhost:8501
Chat
Deploy
Dashboard Controls
Admission Date Range
2025 / 01 / 01 – 2025 / 12 / 31
Patient dataset loaded successfully — 180 patient records
Hospital Key Indicators
Patients
180
Average Age
49.9
Avg Blood Pressure
136.8
Average BMI
26.8
Avg Treatment Cost
₹50,787


Patient Age Distribution
Distribution of Patient Ages
Number of Patients
Age
0
5
10
15
20
20
30
40
50
60
70
80


Dashboard Controls
Admission Date Range
2025 / 01 / 01 – 2025 / 12 / 31
Health Measurement Distributions
Blood Pressure Distribution
Patient Count
Blood Pressure
Cholesterol Distribution
Cholesterol Level
BMI Distribution
BMI
Treatment Cost Distribution
Treatment Cost (₹)


Dashboard Controls
Admission Date Range
2025 / 01 / 01 – 2025 / 12 / 31
Deploy
Health Variables and Treatment Cost Relationships
Age vs Treatment Cost
Treatment Cost (₹)
Age
BMI vs Treatment Cost
BMI
Blood Pressure vs Cholesterol
Cholesterol
Blood Pressure
Age vs BMI


Dashboard Controls
Admission Date Range
2025 / 01 / 01 – 2025 / 12 / 31
Hospital Admissions Over Time
Weekly Hospital Admission Trend
Number of Admissions
Admission Date

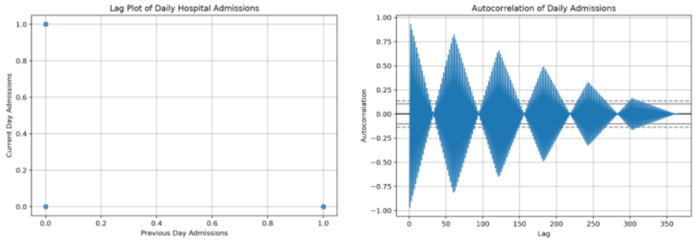

Dashboard Controls
Admission Date Range
2025 / 01 / 01 – 2025 / 12 / 31
Admission Date
Monthly Admission Count
Monthly Hospital Admissions
Number of Admissions
Month

Dashboard Controls

Admission Date Range

2025/01/01 – 2025/12/31

Lag Plot of Hospital Admissions



Dashboard Controls

Admission Date Range

2025/01/01 – 2025/12/31

Health Variable Correlation

	Age	Blood_Pressure	Cholesterol	BMI	Treatment_Cost
Age	1	-0.062	-0.104	-0.114	0.815
Blood_Pressure	-0.062	1	0.03	0.029	0.101
Cholesterol	-0.104	0.03	1	0.097	-0.064
BMI	-0.114	0.029	0.097	1	0.172
Treatment_Cost	0.815	0.101	-0.064	0.172	1

Dashboard Controls

Admission Date Range

2025/01/01 – 2025/12/31

Statistical Summary

	Age	Blood_Pressure	Cholesterol	BMI	Treatment_Cost
count	180	180	180	180	180
mean	49.89	136.82	216.05	26.76	50787.13
std	18.82	19.16	37.72	4.67	10764.3
min	18	105	150	16	28722
25%	33.75	122	183	23.5	43167.5
50%	51	136	218	26.8	50801
75%	65.25	153.25	248	30.22	58672.75
max	80	170	279	36.8	75965

Deploy

Dashboard Controls

Admission Date Range

2025/01/01 – 2025/12/31

Key Hospital Insights

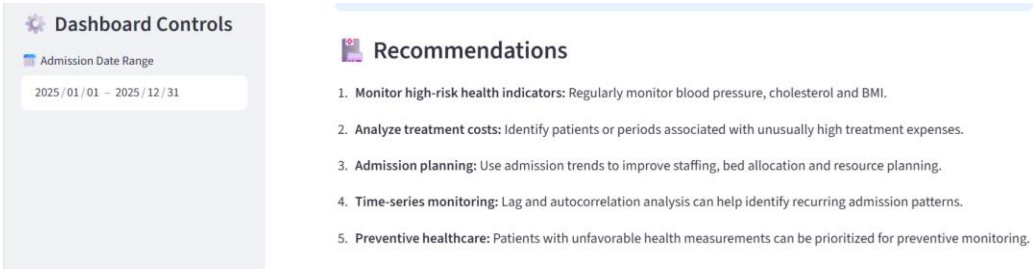
Patient Demographics: The average patient age is **49.9 years**, with patients ranging from **18 to 80 years**.

Blood Pressure: The average blood pressure is **136.8**, indicating that blood-pressure monitoring should receive attention.

BMI: The average BMI is **26.8**. BMI distribution can help identify patients who may require nutritional or lifestyle monitoring.

Treatment Cost: The average treatment cost is **₹50,787**, while the maximum recorded treatment cost is **₹75,965**.

Admission Pattern: The highest number of admissions occurred on **01-01-2025**, with **1 admissions** recorded on that day.



2. **Stock Market Pattern Analysis Dashboard**
   Develop a dashboard using historical stock data containing date,

opening price, closing price, high, low, and trading volume. Visualize price evolution over time, distributions, and relationships between price and volume. Use Pandas plotting, scatter plots, lag plots, autocorrelation plots, and bootstrap plots to investigate trends, dependencies, and uncertainty in stock behavior.

Objectives

1. Import historical stock data using Pandas.
2. Clean and preprocess the dataset.
3. Analyze opening, closing, high, and low prices.
4. Visualize stock-price movement over time.
5. Analyze trading-volume patterns.
6. Study the relationship between stock price and trading volume.
7. Use scatter plots to identify relationships.
8. Use lag plots to identify dependency between consecutive prices.
9. Use autocorrelation plots to detect time-dependent patterns.
10. Apply bootstrap sampling to estimate uncertainty in stock statistics.
11. Develop an interactive dashboard for effective communication.
12. Provide insights useful for understanding stock-market behavior.

Dataset Description

Column	Description
Date	Trading date
Open	Opening price
Close	Closing price
High	Highest price during the day
Low	Lowest price during the day
Volume	Number of shares traded

Deploy

⚙️ **Dashboard Controls**

Select Date Range

2026 / 01 / 02 – 2026 / 06 / 18

Bootstrap Samples

3000

📈 Stock Market Pattern Analysis Dashboard

Historical Stock Market Analysis

This interactive dashboard analyzes stock market behavior using **Pandas and Matplotlib**.

The analysis includes:

- Price evolution
- Price and volume distributions
- Scatter plots
- Lag plot
- Autocorrelation
- Bootstrap analysis
- Correlation analysis

Deploy

⚙️ **Dashboard Controls**

Select Date Range

2026 / 01 / 02 – 2026 / 06 / 18

Bootstrap Samples

3000

Dataset loaded successfully — 120 trading records

📊 Key Market Indicators

Current Price	Average Price	Highest Price	Lowest Price	Average Volume
₹237.57	₹233.45	₹273.72	₹212.35	1,904,142

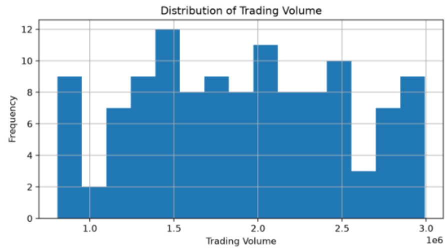
Deploy

📈 Stock Price Evolution

Deploy ⋮

📊 Closing Price Distribution

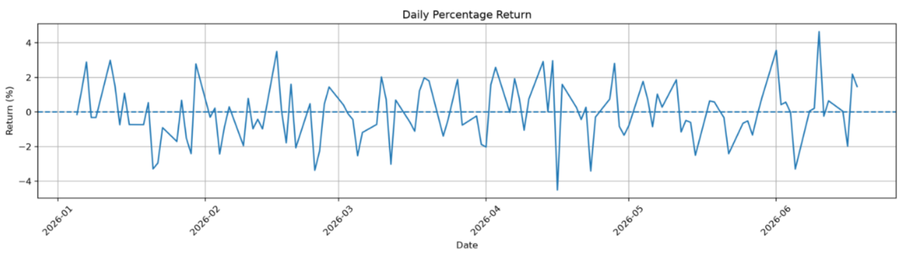
📊 Trading Volume Distribution

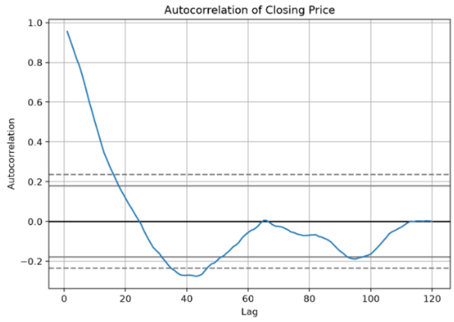


Deploy ⋮

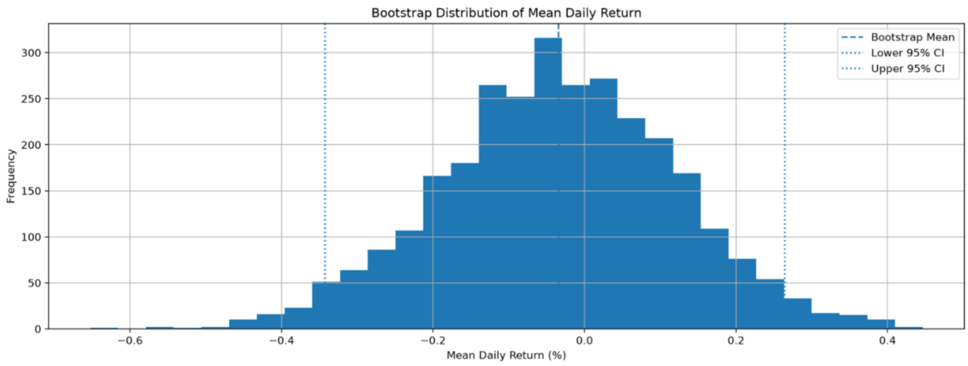
🔵 Relationship Analysis

📉 Daily Return Analysis





🎯 Bootstrap Analysis



Bootstrap Mean

-0.0346%

Lower 95% CI

-0.3424%

Upper 95% CI

0.2642%

🔗 Correlation Analysis

	Open	Close	High	Low	Volume	Return
Open	1	0.991	0.991	0.989	-0.189	0.123
Close	0.991	1	0.99	0.992	-0.176	0.112
High	0.991	0.99	1	0.984	-0.182	0.112
Low	0.989	0.992	0.984	1	-0.163	0.116
Volume	-0.189	-0.176	-0.182	-0.163	1	0.042
Return	0.123	0.112	0.112	0.116	0.042	1

Deploy

📋 Market Summary

Measure	Value
Number of Trading Days	120
Average Closing Price	₹233.45
Maximum Closing Price	₹273.72
Minimum Closing Price	₹212.35
Average Trading Volume	1,904,142
Average Daily Return	-0.0374%
Return Volatility	1.6690%

💡 Key Insights

📉 The stock shows an overall decline of approximately 5.91% during the selected period.

Analysis Recommendations

• Monitor price trends to identify sustained upward or downward movements.

• Examine trading volume along with price changes before making market decisions.

• Use lag and autocorrelation analysis to identify time-dependent behavior.

• Bootstrap confidence intervals provide an estimate of uncertainty in average returns.

• High volatility indicates greater uncertainty and should be considered during risk assessment.